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$Na_3Ca_4(TeO_3)(PO_4)_3$: a new noncentrosymmetric tellurite phosphate with fascinating multimember-ring architectures and intriguing nonlinear optical performance†

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A new functional compound $Na_3Ca_4(TeO_3)(PO_4)_3$ has been successfully synthesized and fully characterized. It is the first example of a tellurite phosphate that exhibits a fascinating multimember-ring configuration. Moreover, a careful theoretical and experimental study also reveals the intriguing nonlinear optical performance of $Na_3Ca_4(TeO_3)(PO_4)_3$.

The exploration of functional compounds with novel structures is of vital importance to enrich the area of structural chemistry.[1–14] Changes in physical and chemical properties originated from novel crystal structures can provide great platforms for the improvement of many important applications, such as porous materials,[15] nonlinear optical (NLO) materials,[16–21] γ-ray detection,[22] energy conversion,[13–25] ionic conductors[26] and solid electrolyte batteries.[27] Interestingly, tellurites exhibit diverse structural topologies due to the nonbonded lone pair electrons on Te^{4+} and the flexible coordination configurations of TeO_x (x = 3, 4, 5).[28] Furthermore, trigonal pyramidal $(TeO_3)^{2-}$, seesaw $(TeO_4)^{4-}$ and square pyramidal $(TeO_5)^{6-}$ units can also connect with other fundamental building blocks by sharing corners and/or edges to construct fascinating framework architectures. Moreover, tellurites have been demonstrated to easily crystallize in the noncentrosymmetric (NCS) space group owing to the second-order Jahn–Teller (SOJT) distortive cation Te^{4+}.[29] And, a NCS structure is the precondition of many significant technological properties such as ferroelectricity, second harmonic generation (SHG) and piezoelectricity. On the other hand, P atoms can only tetrahedrally coordinate to four O atoms to form $(PO_4)^{3-}$ units in phosphates. However, $(PO_4)^{3-}$ groups usually exhibit various polymerizations. Thus, anionic frameworks in phosphates can be classified into orthophosphates (isolated PO_4), pyrophosphates (P_2O_7) and polyphosphates (isolated P_nO_{3n+1} cluster, P_nO_{3n} cyclic rings, $(PO_3)_\infty$ infinite chain).[30] The combination of different P–O groups greatly enriched the structural diversity of phosphates. For instance, $Cs_3Sr_2P_7O_{21}$ contains an isolated P_4O_{12} ring and a one dimensional $[PO_3]_\infty$ chain. $KMg_6(P_2O_7)_2P_3O_{10}$ is composed of isolated P_2O_7 dimers and P_3O_{10} clusters.[30,31]

In addition, the introduction of alkali metals and alkali earth metals into the reaction system can be an effective method to ameliorate the reaction efficiency and reaction path, which is very conducive to achieve the discovery of new crystal structures. Alkali metal and alkali earth metal centred motifs usually exhibit various coordination numbers and these polyhedra can either connect with other fundamental building blocks or interlink together to form multimember rings (MRs) such as 12-NaS_6- and 12-(Zn/Sb)S_4-MRs in $Na_7ZnSb_5S_{12}$.[32] Consequently, combining these building blocks, TeO_x (x = 3, 4, 5) groups, alkali or alkali earth metal centred polyhedra and oxyphosphate units, stands a good chance of generating diverse structural topologies and excellent physical properties. Taking these ideas into consideration, a few tellurite phosphates with multifarious structures have been successfully synthesized, namely $K_2TeP_2O_8$,[33] $BaTeMO_4(PO_4)$ (M = Nb, Ta),[34] $M_2TeMo_2O_6(PO_4)_2$ (M = K, Rb, Cs, Tl),[35] $Ba_2TeO(PO_4)_2$,[29] and $Te_2O(PO_4)_2$.[28] In this work, the first tellurite phosphate containing both alkali metal and alkali earth metals, $Na_3Ca_4(TeO_3)(PO_4)_3$, has been reported. Remarkably, it exhibits intriguing nonlinear optical performance and displays fascinating structural features including 12-CaO_7-MRs, 3-CaO_7-MRs and a $(NaPO_6)_\infty$ tunnel.

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Colourless prism single crystals of the title compound were grown by a high temperature solid state reaction in an open system. Single-crystal structure X-ray diffraction reveals that $Na_3Ca_4(TeO_3)(PO_4)_3$ crystallizes in the hexagonal space group $P6_3$ with the cell parameters of a = 10.7110(3) Å, c = 6.8995(3) Å and Z = 2. To put it in a nutshell, the crystal structure of $Na_3Ca_4(TeO_3)(PO_4)_3$, from outside to inside, is made up of three parts which can be described as 12-CaO_7-MRs, $(NaPO_6)_\infty$ tunnels and $(CaO_3)_\infty$ infinite chains, respectively. And isolated TeO_3 units with stereochemically active 5s lone pairs fill in the voids (Fig. 1a). In an asymmetric unit, there are one Na^+, two Ca^{2+}, one Te^{4+} and five O^{2-} symmetrically independent ions. As illustrated in Fig. 1c and f, Ca1 atoms are coordinated with 7 O atoms to form CaO_7 polyhedra, which are interconnected by sharing the corner oxygen to build the 3-CaO_7-MR with isolated TeO_3 groups filling in it. Note that it is the first example of tellurite phosphate that contains a completely isolated TeO_x group. Furthermore, these 3MRs link with each other by sharing the corner or edge to construct the 12-CaO_7-MRs, which can be considered as the outermost part. With respect to the middle part, NaO_6 units are interlinked to form 3-NaO_6-MRs, which are further put together by PO_4 units and form the $(NaPO_6)_\infty$ tunnel structure (Fig. 1d and e). And, the innermost part features $(CaO_3)_\infty$ infinite chains, which are composed of face-sharing CaO_6 polyhedra (Fig. 1b). The bond lengths of P–O range from 1.480(16) to 1.536(14) Å while those of Te–O are identical to 1.822(13) Å. These lengths are slightly shorter than those of $Ba_2TeO(PO_4)_2$ (1.504(4) to 1.612(4) Å for P–O bonds, 1.917(4) to 2.198(4) Å for Te–O bonds)[29] and comparable to those of $K_2TeP_2O_8$ (1.473(7) to 1.630(12) Å for P–O bonds, 1.779(6) to 2.325(8) Å for Te–O bonds).[28] Besides, the bond valence sums for the cations and anions have been calculated. The results (Na1: 0.724(16); Ca1: 2.098(45); Ca2: 1.612(29); P1: 5.405(139); Te1: 4.581(240); O1: 2.240(174); O2: 2.044(80); O3: 1.870(82); O4: 3.158(87); O5: 2.053(54)) reveal that all atoms are in reasonable oxidation states.

A polycrystalline powder sample of $Na_3Ca_4(TeO_3)(PO_4)_3$ was synthesized by a standard traditional solid state reaction and

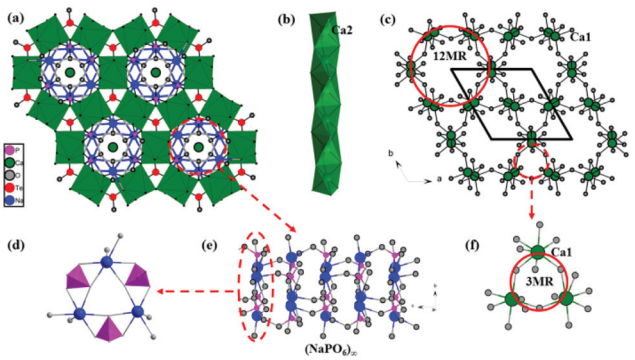

Fig. 1 (a) Crystal structure of $Na_3Ca_4(TeO_3)(PO_4)_3$ along the c-axis, (b) a $(Ca2O_3)_\infty$ infinite chain, (c) the outmost framework is constructed by CaO_7 units. A 12-MR and a 3-MR are indicated with red lines. (d) and (e) the $(NaPO_6)_\infty$ tunnel structure. (f) A 3-CaO_7-MR.

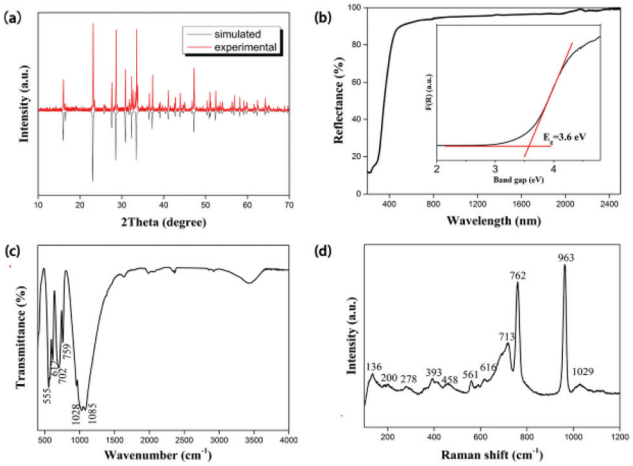

Fig. 2 (a) Powder XRD patterns, (b) UV-vis/NIR diffuse reflectance spectrum, (c) IR spectra, and (d) Raman spectra.

the experimental X-ray diffraction patterns match well with the calculated ones derived from the crystal data, which verifies the sample's purity and the correctness of the crystal structure (Fig. 2a). The UV-vis-NIR spectrum indicates that $Na_3Ca_4(TeO_3)(PO_4)_3$ is transparent down to the ultraviolet spectral region. The optical band gap of $Na_3Ca_4(TeO_3)(PO_4)_3$ was deduced to be 3.6 eV according to the plot of $F(R)$ *versus* $h\nu$ based on the Kubelka–Munk equation (Fig. 2b).[36] Besides, almost no optical absorption emerges below the band gap, which also indirectly demonstrates the purity of the powder sample. The infrared spectrum of $Na_3Ca_4(TeO_3)(PO_4)_3$ is depicted in Fig. 2c. Peaks around 1085 and 1028 cm^{-1} can be attributed to the characteristic vibrations of the PO_4 groups while peaks ranging from 555 to 759 cm^{-1} mainly originate from the Te–O bonds. Raman spectra of $Na_3Ca_4(TeO_3)(PO_4)_3$ were also obtained on the powder sample (Fig. 2d). Peaks located at 963 and 1029 cm^{-1} should be assigned to the P–O bonds and the other two obvious peaks at 762 and 713 cm^{-1} may belong to the Te–O bonds. These values are consistent with the characteristic peaks of other tellurite phosphates.[29,33]

The NLO properties of $Na_3Ca_4(TeO_3)(PO_4)_3$ are evaluated by a powder SHG method.[37] The powder samples and reference KH_2PO_4 (KDP) crystals were thoroughly ground and sieved into five groups of distinct particle size ranges, which were then irradiated under a 1064 nm laser. As illustrated in Fig. 3a, the SHG intensities increase gradually with the increase of the particle size and finally form a plateau. This kind of curve indicates that $Na_3Ca_4(TeO_3)(PO_4)_3$ displays a typical type-I phase-matching behavior, based on the rule proposed by Kurtz and Perry.[37] It is worth noting that good phase-matchability is favorable for the application of NLO materials. Moreover, the powder SHG signals are about 0.2 times that of the benchmark crystal KDP when compared at the particle size of 150–200 mm (Fig. 3b). According to the anionic group theory, a large NLO response can be achieved when favorable NLO-active building blocks occur in high density and in coparallel alignment.[38]

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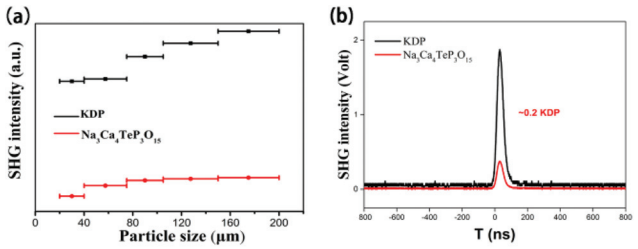

Fig. 3 (a) Phase-matching curve, (b) oscilloscope traces of SHG signals of $Na_3Ca_4(TeO_3)(PO_4)_3$ and KDP (reference) in the particle size of 150–200 μm.

Structurally, the relatively weak NLO response of $Na_3Ca_4(TeO_3)(PO_4)_3$ mainly stems from the unfavorable alignments of TeO_3 and PO_4 groups which tilt to each other around the 6_3 screw axis.

To further explore the relationship between structure and performance, their electronic structures are investigated by density functional theory (DFT).[39–41] The electronic band structure along the highly symmetrical path indicates that $Na_3Ca_4(TeO_3)(PO_4)_3$ is a direct transition compound with the calculated band gap of 3.6 eV, which is highly consistent with the experimental value (Fig. 4b). The density of states (DOS) of the title compound is depicted in Fig. 4a. Strong hybridization between Te and O, as well as P and O, can be easily found in the valence bands (VB) and conduction bands (CB), which indicates the covalent characteristics. Clearly, the top of the VB is mainly composed of O 2p orbitals with unneglectable amounts of Te 5s, 5p and P 3s, 3p orbitals. Besides, Te 5p orbitals account for the major part of the bottom of the CB. Since materials' optical properties principally stem from the electronic transitions between the VB and CB states in the vicinity of the band gap, Te–O groups are the main contributors to the optical response of $Na_3Ca_4(TeO_3)(PO_4)_3$. Furthermore, the refractive indices are also calculated and the birefringence of $Na_3Ca_4(TeO_3)(PO_4)_3$ at 1064 nm is 0.05, which is larger than those of most of the phosphates. The enhanced birefringence may originate from the introduction of the stereochemically active lone pair (SCALP) cation Te^{4+}, which possesses a local acentric coordination environment and a larger microscopic susceptibility and thus gives rise to strengthened structural anisotropy. The above phenomenon has also been proven in other compounds with lone pair electrons.[32,42] And, the electron localization function (ELF) maps (Fig. S2†) prove the existence of lone pair electrons around the Te atoms. Note that such a suitable birefringence is desirable for the phase-matching behavior, which is one of the prerequisites of a practically usable NLO crystal. Moreover, the effect of SCALP cations on increasing birefringence may shed light on a new path to improve the phase-matchability of phosphates.

To sum up, a new tellurite phosphate $Na_3Ca_4(TeO_3)(PO_4)_3$ has been successfully synthesized and fully characterized. It crystallizes in the noncentrosymmetric space group $P6_3$ and features a fascinating multimember-ring configuration. Interestingly, from outside to inside, it is composed of three parts which can be depicted as 12-CaO_7-MRs, $(NaPO_6)_\infty$ tunnels and $(CaO_3)_\infty$ infinite chains. The introduction of Te^{4+} with a local acentric coordination environment and a larger microscopic susceptibility strengthens the anisotropy of the whole structure which is favorable for NLO applications. In addition, $Na_3Ca_4(TeO_3)(PO_4)_3$ is transparent down to the ultraviolet region with a bandgap of 3.6 eV. A combined theoretical and experimental study illustrates that $Na_3Ca_4(TeO_3)(PO_4)_3$ shows good phase-matchability with a moderate SHG response at 1064 nm that is approximately 0.2 times that of KDP.

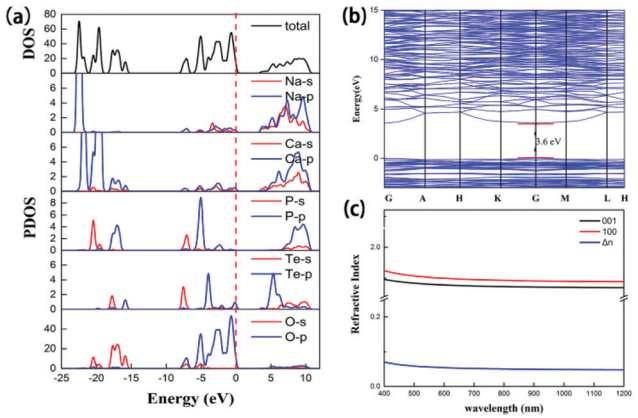

Fig. 4 (a) DOS of $Na_3Ca_4(TeO_3)(PO_4)_3$, (b) electron band structure of $Na_3Ca_4(TeO_3)(PO_4)_3$, (c) dispersion curves of the refractive indexes and birefringence of $Na_3Ca_4(TeO_3)(PO_4)_3$.

Conflicts of interest

There are no conflicts to declare.

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